

Lecture 14: Fully Bayesian Models

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14.1 Power Ratings Are Not Just Rankings

Suppose Dallas's estimated strength is two points above Philadelphia's. Here strength is measured in expected point differential relative to league average. Which team should be favored on a neutral field?

A ranking gives one answer. A decision-maker usually needs more:

- What is $\mathbb{P}(\beta_{\text{DAL}} > \beta_{\text{PHI}} \mid \text{data})$?
- What is $\mathbb{P}(\text{Dallas is a top-five team} \mid \text{data})$?
- What is the predictive probability Dallas wins one future game?
- How much uncertainty comes from team quality versus game randomness?

A fully Bayesian model assigns probability distributions to unknown quantities, updates those distributions with observed data, and propagates the remaining uncertainty into decisions and predictions.

14.2 Observed Scores and Unknown Strengths

Let y_i be home score minus away score in game i . Let $H(i)$ and $A(i)$ identify the home and away teams. A simple one-season power-rating model could look like

$$y_i \sim \mathcal{N}(\beta_0 + \beta_{H(i)} - \beta_{A(i)}, \sigma_{\text{game}}^2).$$

Each component has a sports interpretation:

- β_0 : league-wide home-field advantage.
- β_j : unobserved strength of team j , in points above average.
- σ_{game} : game-to-game football randomness.

Games identify strength differences such as $\beta_{H(i)} - \beta_{A(i)}$. To interpret team strength as points above league average, center the team effects:

$$\sum_j \beta_j = 0.$$

Without a constraint or equivalent parameterization, adding the same constant to every team strength would not change the likelihood.

Game	Home	Away	Week	Season	Home margin
2018_01_ATL_PHI	PHI	ATL	1	2018	6
2018_01_BUF_BAL	BAL	BUF	1	2018	44
2018_01_CHI_GB	GB	CHI	1	2018	1
2018_01_CIN_IND	IND	CIN	1	2018	-11
2018_01_DAL_CAR	CAR	DAL	1	2018	8
2018_01_HOU_NE	NE	HOU	1	2018	7
2018_01_JAX_NYG	NYG	JAX	1	2018	-5

Table 14.1: Sample of observed NFL games used to fit the power-rating model.

14.3 Priors on Football Quantities

In Lecture 11, prior information was encoded as fake wins and fake losses for one probability. Here, priors encode plausible values for several football quantities:

$$\begin{aligned}\beta_0 &\sim \mathcal{N}(0, 5^2), \\ \beta_j \mid \sigma_{\text{team}} &\sim \mathcal{N}(0, \sigma_{\text{team}}^2), \\ \sigma_{\text{game}}, \sigma_{\text{team}} &\sim \text{half-Normal}(0, 10^2).\end{aligned}$$

These priors have direct interpretations:

- the prior for β_0 describes plausible home-field advantages;
- the prior for β_j says teams start near league average before their games are observed;
- σ_{game} describes how noisy individual football games are;
- σ_{team} describes how spread out team strengths are across the league.

14.4 Pooling Comes From the Prior

The team-strength prior creates partial pooling. A team with limited or noisy evidence is pulled more toward league average. A team with strong evidence can move farther away from zero.

The amount of pooling is controlled by σ_{team} :

- small σ_{team} : teams are believed to be similar, so shrinkage is stronger;
- large σ_{team} : teams can differ more, so shrinkage is weaker.

This connects directly to shrinkage estimation. Lecture 13 chose or estimated a shrinkage rule for many noisy estimates. A fully Bayesian model puts a probability distribution on the quantity that controls the shrinkage.

14.5 Priors on Priors

Empirical Bayes estimates prior hyperparameters and then treats them as fixed. A fully Bayesian model keeps those quantities uncertain. The data update beliefs about the teams, but also about how different teams tend to be.

For multiple seasons, let $\beta_{j,s}$ be team j 's strength in season s . A dynamic model can share information across seasons:

$$\begin{aligned}\beta_{j,1} &\sim \mathcal{N}(0, \sigma_{\text{team}}^2), \\ \beta_{j,s} &\sim \mathcal{N}(\gamma\beta_{j,s-1}, \sigma_{\text{season}}^2), \quad s > 1, \\ \gamma &\sim \text{Uniform}(0, 1), \\ \sigma_{\text{season}} &\sim \text{half-Normal}(0, 10^2).\end{aligned}$$

Here γ describes persistence and σ_{season} describes how much team quality changes between seasons.

The posterior combines the score model, the priors on team strengths, and the priors on the quantities controlling those priors:

$$\mathbb{P}(\boldsymbol{\beta}, \beta_0, \gamma, \boldsymbol{\sigma} \mid \mathbf{y}) \propto \mathbb{P}(\mathbf{y} \mid \boldsymbol{\beta}, \beta_0, \sigma_{\text{game}}) \mathbb{P}(\boldsymbol{\beta} \mid \gamma, \sigma_{\text{team}}, \sigma_{\text{season}}) \mathbb{P}(\beta_0, \gamma, \boldsymbol{\sigma}).$$

14.6 Posterior Draws Are Plausible Football Worlds

For a model this large, the posterior is not usually available as a simple named distribution. Software such as Stan uses Markov chain Monte Carlo to generate approximate posterior draws:

$$\Theta^{(1)}, \Theta^{(2)}, \dots, \Theta^{(M)} \sim \mathbb{P}(\Theta \mid \mathbf{y}),$$

where Θ contains all model parameters.

Each draw is one plausible football world after seeing the data. It contains one home-field advantage, one set of team strengths, one value for game randomness, one value for league-wide team spread, and, in the dynamic model, one value for season-to-season persistence.

The figures below use a conjugate Gaussian power-rating version with fixed variance parameters, so the posterior can be drawn directly. The same draw-by-draw logic applies to the full dynamic model, where MCMC replaces the direct Gaussian calculation.

14.7 Count Draws to Answer Questions

Posterior draws turn probability questions into counting problems. For Dallas and Philadelphia,

$$\mathbb{P}(\beta_{\text{DAL}} > \beta_{\text{PHI}} \mid \mathbf{y}) \approx \frac{1}{M} \sum_{m=1}^M \mathbb{1}\{\beta_{\text{DAL}}^{(m)} > \beta_{\text{PHI}}^{(m)}\}.$$

Ranking teams by strength within each draw gives each team's posterior probability of having top-five strength:

$$\mathbb{P}(\text{Dallas is top five} \mid \mathbf{y}) \approx \frac{1}{M} \sum_{m=1}^M \mathbb{1}\{\text{Dallas ranks in the top five in draw } m\}.$$

Posterior Summaries

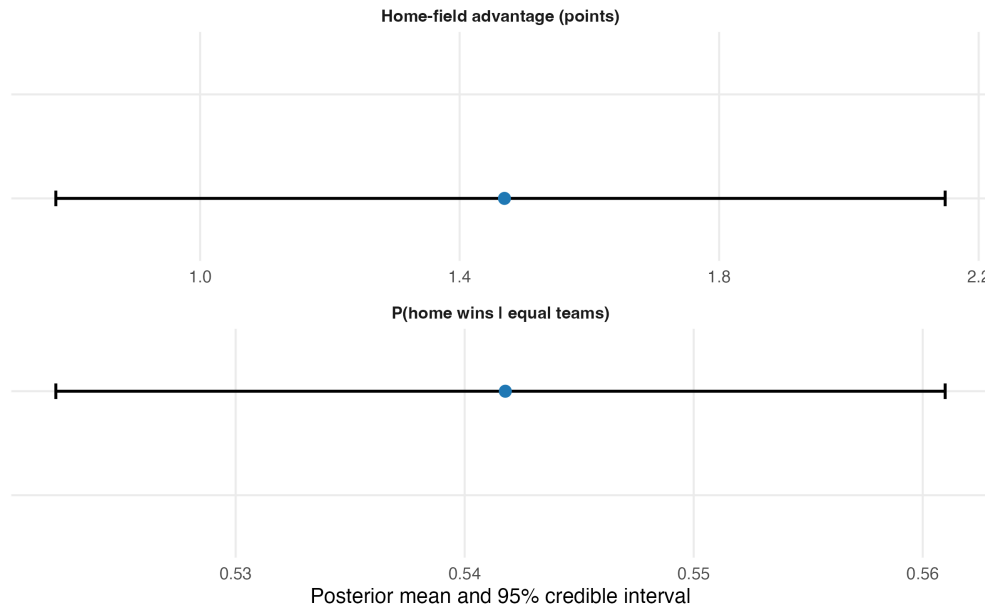


Figure 14.1: Posterior summaries for home-field advantage and equal-team home-win probability.

A future neutral-site score differential can also be simulated from each draw:

$$y_{\text{future}}^{(m)} \sim \mathcal{N}\left(\beta_{\text{DAL}}^{(m)} - \beta_{\text{PHI}}^{(m)}, \left(\sigma_{\text{game}}^{(m)}\right)^2\right).$$

The fraction with $y_{\text{future}}^{(m)} > 0$ estimates the predictive probability that Dallas wins one game.

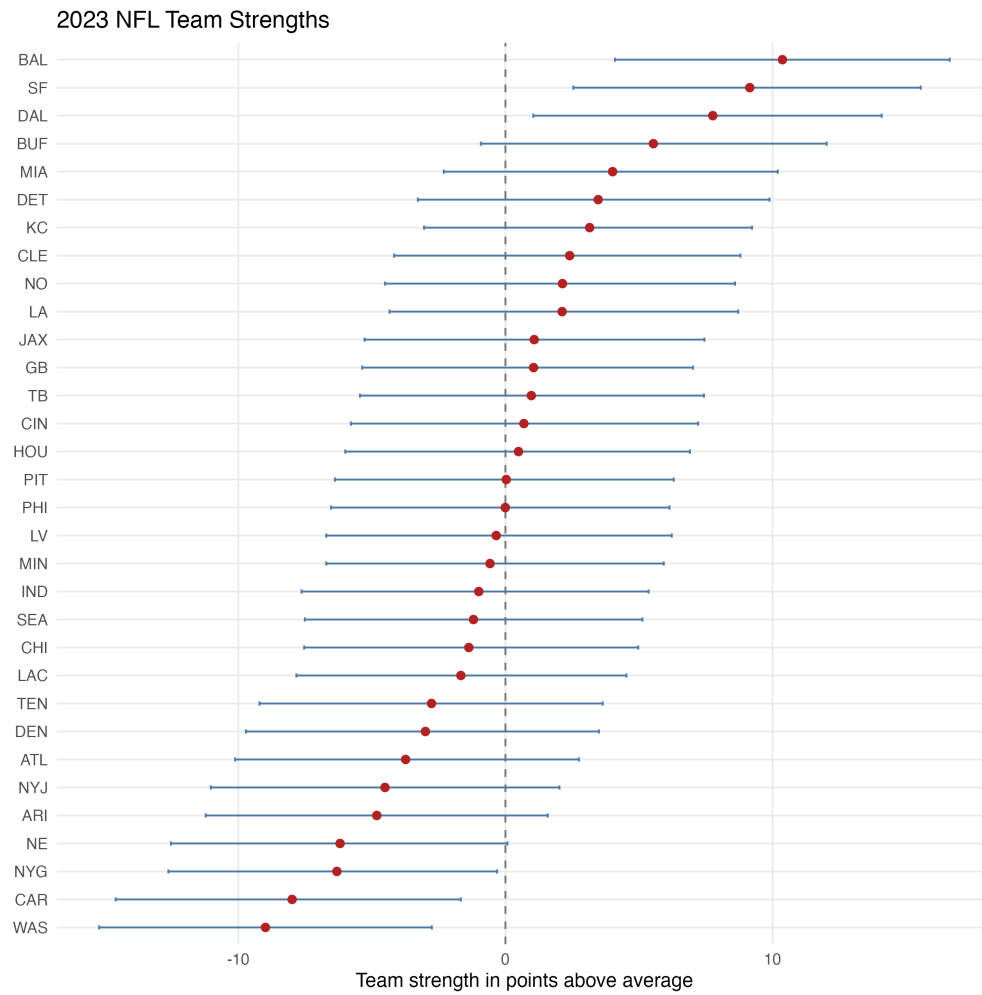


Figure 14.2: Posterior means and 95% credible intervals for team strength.

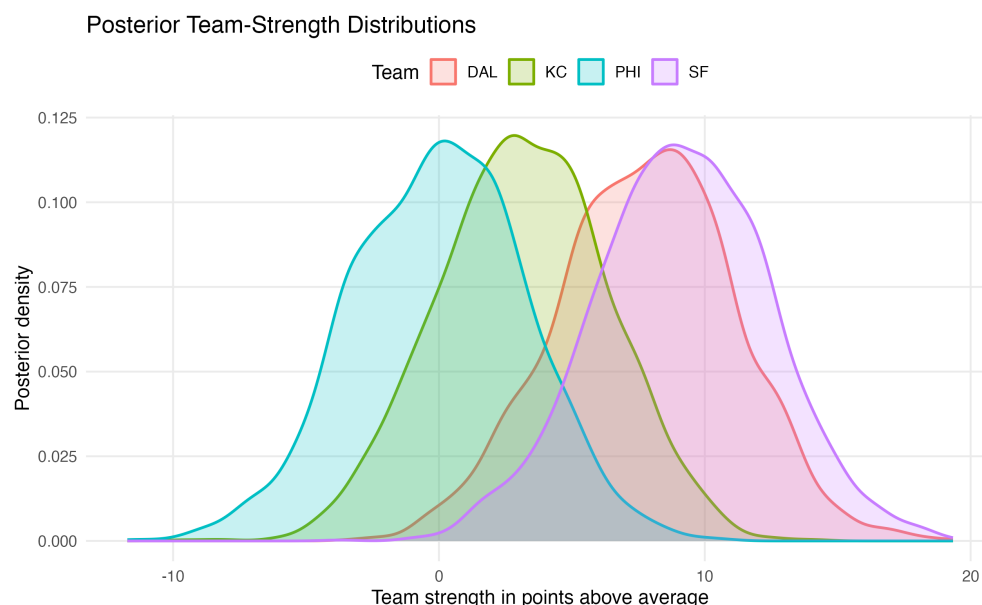


Figure 14.3: Posterior team-strength distributions for four teams.

14.8 Parameter and Predictive Uncertainty

The probability that Dallas is stronger than Philadelphia is not the same as the probability Dallas wins one game. The first question concerns parameters. The second question concerns a future outcome.

A 95% credible interval for team strength uses the 2.5% and 97.5% posterior quantiles. It describes uncertainty about an unobserved parameter.

A 95% posterior predictive interval for a future score differential also includes game-level randomness. It will generally be much wider:

$$\text{future uncertainty} = \text{parameter uncertainty} + \text{outcome randomness}.$$

Confusing these intervals leads to overconfidence. Knowing team quality precisely would not make football outcomes deterministic.

14.9 Check the Priors and the Fitted Model

Once priors become part of the model, they become part of what must be checked. A complicated posterior is not evidence that the model is good.

14.9.1 Prior Predictive Checks

Before fitting, simulate parameters from the priors and games from the model. Do these priors generate plausible team strengths, home advantages, and score differentials? If not, revise the priors.

14.9.2 Posterior Predictive Checks

After fitting, simulate replicated seasons from the posterior. Compare simulated and observed quantities such as:

- distribution of score differentials;
- rate of close games and blowouts;
- home win rate;
- team-level residual patterns;
- out-of-sample calibration and prediction error.

These checks ask whether the model can reproduce important features of the data, not merely whether computation completed.

14.9.3 Model Limitations

Model criticism asks which omitted features matter for the intended decision and whether adding them improves calibrated out-of-sample predictions.

14.10 What Fully Bayesian Models Buy

Approach	Treated as known	Uncertainty kept	Answer
Point ratings	One fitted rating per team	Usually none in the rating table	Rankings; spreads
Empirical Bayes	Estimated prior center and spread	Team effects given the fitted prior	Shrunk estimates
Fully Bayesian	Model choices and prior families	Strengths, home field, game noise, shrinkage, persistence	Probabilities

Fully Bayesian modeling does not remove modeling judgment. It makes the judgment explicit, updates uncertain quantities together, and carries that uncertainty into the final decision.

14.11 Takeaways

- A fully Bayesian model assigns uncertainty to unknown quantities at every layer.
- Priors can encode home-field advantage, team strength, game randomness, shrinkage strength, and season-to-season persistence.
- Posterior draws turn strength comparisons, rankings, and future predictions into direct calculations.
- Credible intervals describe parameters; posterior predictive intervals describe future outcomes.
- Prior and posterior predictive checks reveal implausible assumptions and systematic model errors.

Appendix

A Computing the Posterior

Posterior computation for a full hierarchical model requires four steps:

1. encode games, team indices, and seasons as data;
2. write the likelihood, hierarchical priors, and centering constraint in Stan;
3. run multiple Markov chains to explore the posterior;
4. extract draws and translate them into sports questions.

Posterior draws are useful only if the sampler explored the posterior adequately. At minimum, report:

- $\hat{R}$ values close to 1;
- adequate effective sample sizes;
- no divergent transitions;
- well-mixed trace plots from multiple chains.

These diagnose computation. Posterior predictive checks diagnose the model. Passing one does not imply passing the other.

B Bayesian Models and Regularization

Bayesian modeling and regularization often produce the same point estimate. In Gaussian regression,

$$\begin{aligned}\mathbf{y} \mid \boldsymbol{\beta} &\sim \mathcal{N}(X\boldsymbol{\beta}, \sigma^2 I), \\ \boldsymbol{\beta} &\sim \mathcal{N}(0, \sigma^2 \lambda^{-1} I),\end{aligned}$$

the posterior mean is

$$\mathbb{E}[\boldsymbol{\beta} \mid \mathbf{y}] = (X^T X + \lambda I)^{-1} X^T \mathbf{y},$$

the ridge-regression estimate. The Bayesian model additionally supplies a posterior distribution and posterior predictive distribution.

C Suggested Reading

- Mark Glickman and Hal Stern, “A State-Space Model for National Football League Scores.”
- Andrew Gelman et al., *Bayesian Data Analysis*, on posterior computation, diagnostics, and predictive checking.
- Shachi Deshpande and Abraham Wyner, “A Hierarchical Bayesian Model of Pitch Framing.”